Schedule

Practice problems/reading assignments on HT: on LearnHub

Lab 5: on LearnHub, today

Case Exam 1: Saturday, April 19th at 18:30-21:00 in labs SNAB149 and SNAB242

- Seating charts will be announced
- Only lab computers can be used
- Safe Exam Browser; Phyton, Excel, common report area

Will have access to LearnHub material

Midterm: Friday, April 25th at 19:00-21:00, on LearnHub, multiple choice, in labs SNAB149 and SNAB220

- Safe Exam Browser, only Excel can be used
- Seating charts will be announced

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Agenda

Significance testing: short review

Class exercise

HT/ST for normal random variable with small sample size

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Hypothesis/Significance Tests

HYPOTHESIS	SIGNIFICANCE
Parameter of interest and its estimator	Parameter of interest and its estimator
Null/alternative hypotheses	Null/alternative hypotheses
Test statistic and its distribution (assuming the null hypothesis is true)	Test statistic and its distribution (assuming the null hypothesis is true)
Rejection region	Observe the sample and calculate the test statistics or the estimator
Observe the sample and calculate the test statistics or the estimator	Calculate the corresponding p-value
Decide	Decide

Another approach to hypothesis testing: *P*-values

Hypothesis testing:

$$\alpha \rightarrow \underbrace{\text{Rejection Region}}_{\text{computation of the critical value, k}} \rightarrow \underbrace{\text{conclusion}}_{\text{comparison of } (\bar{x} \text{ and k}) \text{ or } (z \text{ and } z_\alpha)}$$

Significance testing:

$$\bar{x} \rightarrow \underbrace{\text{P}\{\text{type I error when } H_0 \text{ is rejected with } k = \bar{x}\}}_{\text{computation of the P-value, P}} \rightarrow \underbrace{\text{conclusion}}_{\text{comparison of P-value and } \alpha}$$

P-values

Definition: The p-value (or observed significance level) is the smallest level of significance at which H_0 would be rejected when a specified test procedure is used on a given data set. Once the p-value has been determined, the conclusion at any particular level α results from comparing the P-value to α:

1) P-value $\leq \alpha$ → reject H_0 at level α

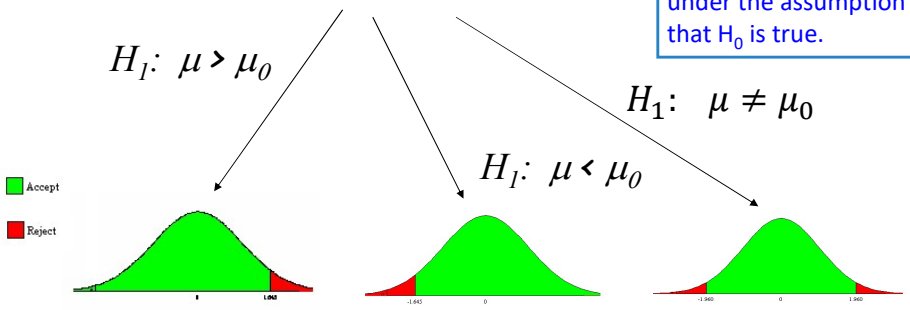
2) P-value $> \alpha$ → cannot reject H_0 at level α

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Tests about a population mean of a normal r.v. with known variance (Hypothesis Testing)

H_0: $\mu = \mu_0$ Test statistics: $Z = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$

Test statistics is given under the assumption that H_0 is true.

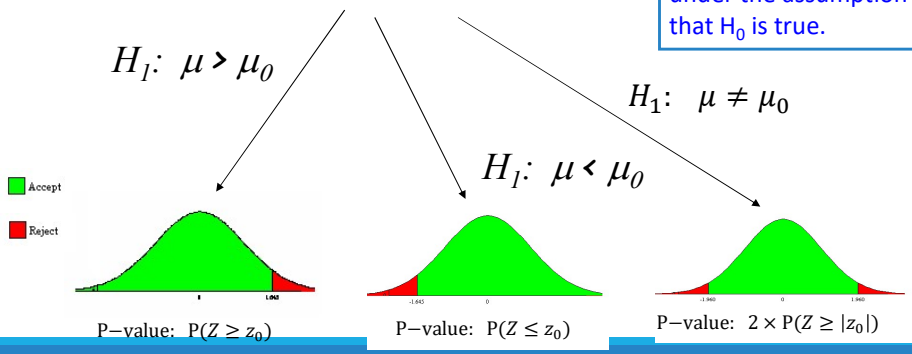


6

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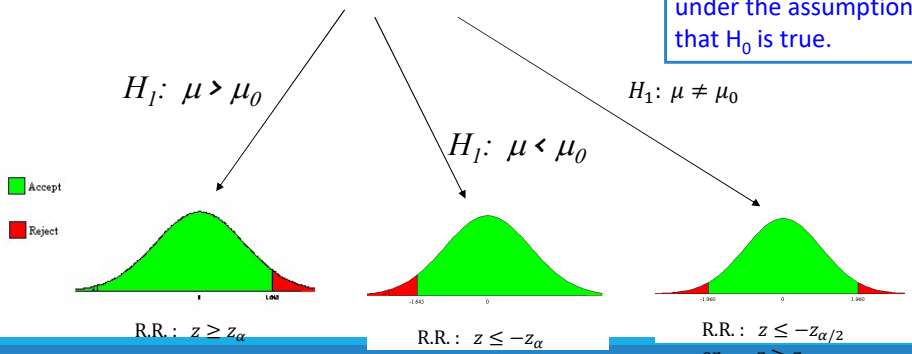
P−value: $P(Z \geq z_0)$ P−value: $P(Z \leq z_0)$ P−value: $2 \times P(Z \geq |z_0|)$

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Tests about a population mean of a r.v. with large sample size

H_0: $\mu = \mu_0$ Test statistics: $Z = \dfrac{\bar{X} - \mu_0}{S/\sqrt{n}}$

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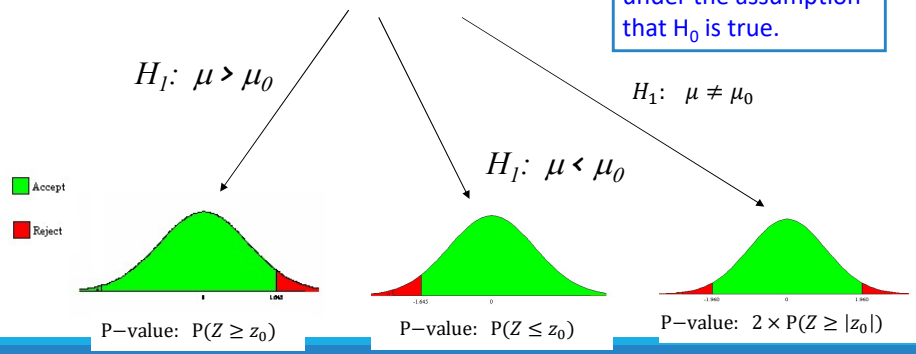
R.R. : $z \geq z_\alpha$ R.R. : $z \leq -z_\alpha$ R.R. : $z \leq -z_{\alpha/2}$ or $z \geq z_{\alpha/2}$

8

Tests about a population mean of a r.v. with large sample size (Significance Testing)

H_0: $\mu = \mu_0$ Test statistics: $z_0 = \dfrac{\bar{x} - \mu_0}{S/\sqrt{n}}$

Test statistics is given under the assumption that H_0 is true.



P−value: $P(Z \geq z_0)$ P−value: $P(Z \leq z_0)$ P−value: $2 \times P(Z \geq |z_0|)$

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Class exercise

The target thickness for silicon wafers used in a certain type of integrated circuit is 245 mm. A sample of 50 wafers is obtained and the thickness of each one is determined, resulting in a sample mean of 246.18 mm. and a sample standard deviation of 3.6 mm. Does this data suggest that true average wafer thickness is something other than the target value?

- Formulate the hypothesis test.
- Find the p-value.
- What is your conclusion for α=1%? How about α=5%?

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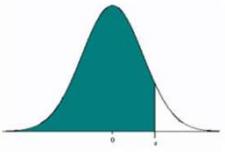


Table of Standard Normal Probabilities for Positive Z-scores

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

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Hypothesis testing versus CIs

H_0: $\mu = \mu_0$
H_a: $\mu \neq \mu_0$
at significance level α

If μ_0 is in (l,u), do we reject H_0? Why?

2-sided confidence interval (l,u) at 100(1-α)%

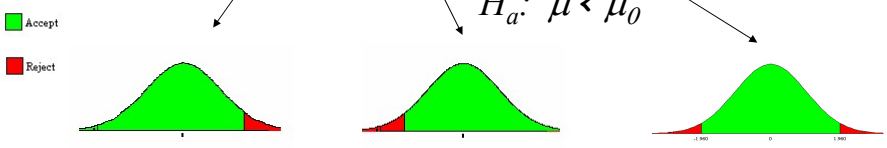
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Tests about a population mean of a normal r.v. with unknown variance and small sample size

H_0: $\mu = \mu_0$ **Test statistics:** $T = \dfrac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1}$

Test statistics is given under the assumption that H_0 is true.

H_a: $\mu > \mu_0$ H_a: $\mu < \mu_0$ H_a: $\mu \neq \mu_0$



R.R.: $t \geq t_\alpha$ R.R.: $t \leq -t_\alpha$ R.R.: $t \leq -t_{\alpha/2}$ or $t \geq t_{\alpha/2}$

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Example

Service times of incoming calls in an airline reservation system: $S_i \sim N(\mu, \sigma)$ with σ is unknown.

Average service time in the biggest 10 airline companies is reported as 100 seconds.

$$H_0: \quad \mu = 100$$

If the average is 100 or less, there will be no change in the system.

$$H_a: \quad \mu > 100$$

If the average has increased above 100, more capacity is needed, i.e., number of servers in the call center should be increased.

$\bar{x} = 111.38$ $n = 15$ $\alpha = 0.05$ $s = 19.28$

Is more capacity needed?

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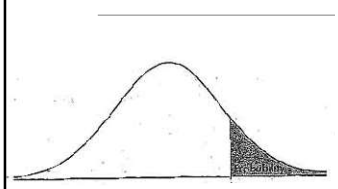


TABLE B: *t*-DISTRIBUTION CRITICAL VALUES

	Tail probability p											
df	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
1	1.000	1.376	1.963	3.078	6.314	12.71	15.89	31.82	63.66	127.3	318.3	636.6
2	.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	31.60
3	.765	.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	12.92
4	.741	.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	.727	.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	.718	.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	.711	.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	.706	.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	.703	.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	.700	.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	.697	.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	.695	.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	.694	.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	.692	.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	.691	.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	.690	.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	.689	.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	.688	.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.611	3.922
19	.688	.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	.687	.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	.686	.859	1.063	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	.686	.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792
23	.685	.858	1.060	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768
24	.685	.857	1.059	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745
25	.684	.856	1.058	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725
26	.684	.856	1.058	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707
27	.684	.855	1.057	1.314	1.703	2.052	2.158	2.473	2.771	3.057	3.421	3.690
28	.683	.855	1.056	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674
29	.683	.854	1.055	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.659
30	.683	.854	1.055	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646
40	.681	.851	1.050	1.303	1.684	2.021	2.123	2.423	2.704	2.971	3.307	3.551
50	.679	.849	1.047	1.299	1.676	2.009	2.109	2.403	2.678	2.937	3.261	3.496
60	.679	.848	1.045	1.296	1.671	2.000	2.099	2.390	2.660	2.915	3.232	3.460
80	.678	.846	1.043	1.292	1.664	1.990	2.088	2.374	2.639	2.887	3.195	3.416
100	.677	.845	1.042	1.290	1.660	1.984	2.081	2.364	2.626	2.871	3.174	3.390
1000	.675	.842	1.037	1.282	1.646	1.962	2.056	2.330	2.581	2.813	3.098	3.300
∞	.674	.841	1.036	1.282	1.645	1.960	2.054	2.326	2.576	2.807	3.091	3.291

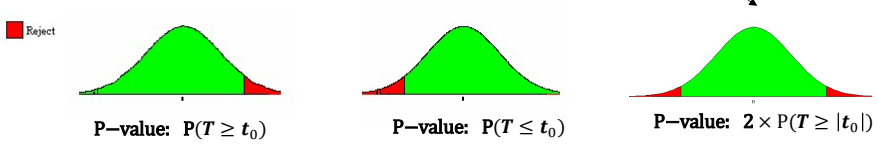
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Tests about a population mean of a normal r.v. with unknown variance and small sample size

H_0: $\mu = \mu_0$ **Test statistics:** $t_0 = \dfrac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1}$

Test statistics is given under the assumption that H_0 is true.

H_a: $\mu > \mu_0$ H_a: $\mu < \mu_0$ H_a: $\mu \neq \mu_0$



P–value: $P(T \geq t_0)$ **P–value:** $P(T \leq t_0)$ **P–value:** $2 \times P(T \geq |t_0|)$

Find bounds on the p-value using the t-table.

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Example

Service times of incoming calls in an airline reservation system: $S_i \sim N(\mu, \sigma)$ with unknown σ.

H_0: $\mu = 100$
H_a: $\mu > 100$

$\overline{x} = 111.38$
$s = 19.28$
$n = 15$

Find bounds on the p-value. Is more capacity needed at α=0.01? α=0.05?

0.01 < p-value < 0.02

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Power and sample size in a t-test

Complicated → You can find appropriate functions in Python, but we will not use them.

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Large-sample tests for a population proportion

X : # of successes in a random sample with n << population size

p : proportion of 'success' in a population

X~bin(n,p)

- For large n with np ≥ 10 and n(1-p) ≥ 10, X is approximately normal with $E(X) = np$, & $V(X) = np(1-p)$.
- $\hat{p} = X/n$ is unbiased since $E(\hat{p}) = p$, has approximately a normal distribution, and its standard deviation is $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$.

$H_0: p = p_0$

true only if $np_0 \geq 10$ & $n(1-p_0) \geq 10$

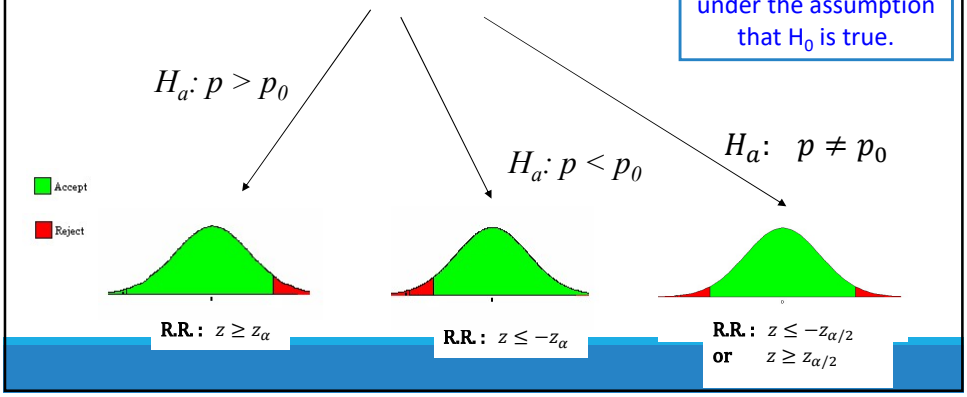
When H_0 is true, $Z = \dfrac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}} \sim N(0,1)$

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Test statistics, H_a & RR for population proportion *p*

$H_0: p = p_0$ **Test statistics:** $Z = \dfrac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$

Test statistics is given under the assumption that H_0 is true.

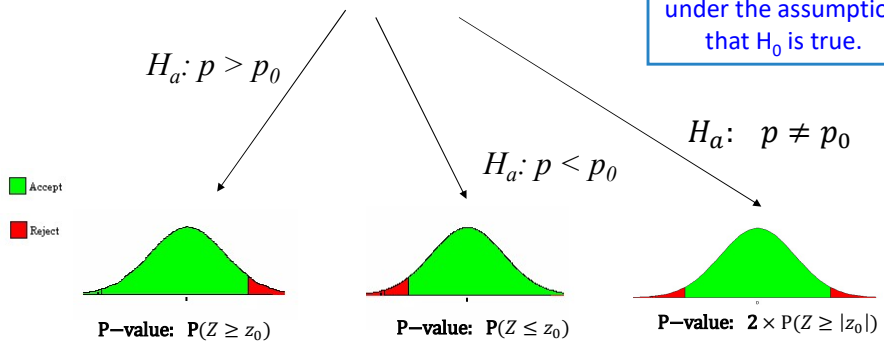


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Test statistics, H_a & RR for population proportion p

$H_0: p = p_0$ **Test statistics:** $z_0 = \dfrac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$

Test statistics is given under the assumption that H_0 is true.



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Hypothesis Testing for two populations

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Tests about means of two normal populations with known variance

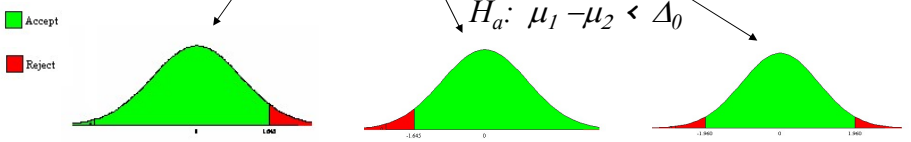
H_0: $\mu_1 - \mu_2 = \Delta_0$

Test statistics: $Z = \dfrac{\bar{X} - \bar{Y} - \Delta_0}{\sqrt{\dfrac{\sigma_1^2}{m} + \dfrac{\sigma_2^2}{n}}}$

H_a: $\mu_1 - \mu_2 > \Delta_0$

H_a: $\mu_1 - \mu_2 \neq \Delta_0$

H_a: $\mu_1 - \mu_2 < \Delta_0$



R.R.: $z \geq z_\alpha$

R.R.: $z \leq -z_\alpha$

R.R.: $z \leq -z_{\alpha/2}$ or $z \geq z_{\alpha/2}$

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Tests about means of two populations with large sample size

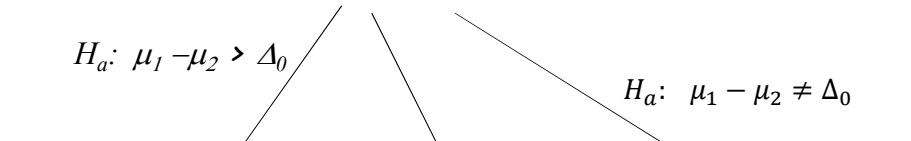
H_0: $\mu_1 - \mu_2 = \Delta_0$

Test statistics: $Z = \dfrac{\bar{X} - \bar{Y} - \Delta_0}{\sqrt{\dfrac{S_1^2}{m} + \dfrac{S_2^2}{n}}}$

H_a: $\mu_1 - \mu_2 > \Delta_0$

H_a: $\mu_1 - \mu_2 \neq \Delta_0$

H_a: $\mu_1 - \mu_2 < \Delta_0$



R.R.: $z \geq z_\alpha$

R.R.: $z \leq -z_\alpha$

R.R.: $z \leq -z_{\alpha/2}$ or $z \geq z_{\alpha/2}$

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Tests about means of two normal populations with unknown variance

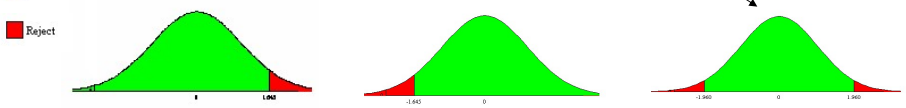
H_0: $\mu_1 - \mu_2 = \Delta_0$

Test statistics: $t = \dfrac{\bar{x} - \bar{y} - \Delta_0}{\sqrt{\dfrac{s_1^2}{m} + \dfrac{s_2^2}{n}}}$

H_a: $\mu_1 - \mu_2 > \Delta_0$

H_a: $\mu_1 - \mu_2 < \Delta_0$

H_a: $\mu_1 - \mu_2 \neq \Delta_0$



R.R.: $t \geq t_{\alpha,\nu}$

R.R.: $t \leq -t_{\alpha,\nu}$

R.R.: $t \leq -t_{\alpha/2,\nu}$ **or** $t \geq t_{\alpha/2,\nu}$

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Example

The production manager wants to compare the average weights of popcorns from two production lines. Experience shows that the weights of popcorns can be assumed to be normal. He would like to know whether the true mean of the weights of popcorns from two production lines are different than each other. He obtained the following data:

	sample size	sample mean	σ pop. std. deviation
Line 1	12	5.2	0.5
Line 2	16	5.6	0.6

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Example

	sample size	sample mean	σ pop. std. deviation
Line 1	12	5.2	0.5
Line 2	16	5.6	0.6

The production manager wants to compare the average weights of popcorns from two production lines. Experience shows that the weights of popcorns can be assumed to be normal. He would like to know whether the true mean of the weights of popcorns from two production lines are different than each other.

Formulate the appropriate hypothesis testing.

$$H_0: \ \mu_1 = \mu_2$$
$$H_a: \ \mu_1 \neq \mu_2$$

Example

	sample size	sample mean	σ pop. std. deviation
Line 1	12	5.2	0.5
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The production manager wants to compare the average weights of popcorns from two production lines. Experience shows that the weights of popcorns can be assumed to be normal. He would like to know whether the true mean of the weights of popcorns from two production lines are different than each other.

$$H_0: \ \mu_1 = \mu_2$$
$$H_a: \ \mu_1 \neq \mu_2$$

Let α=0.05. Can you reject H_0?

Example

	sample size	sample mean	σ pop. std. deviation
Line 1	12	5.2	0.5
Line 2	16	5.6	0.6

The production manager wants to compare the average weights of popcorns from two production lines. Experience shows that the weights of popcorns can be assumed to be normal. He would like to know whether the true mean of the weights of popcorns from two production lines are different than each other.

$$H_0: \ \mu_1 = \mu_2$$
$$H_a: \ \mu_1 \neq \mu_2$$

What is the p-value of the test for the given data?

Example

	sample size	sample mean	σ pop. std. deviation
Line 1	12	5.2	0.5
Line 2	16	5.6	0.6

The production manager wants to compare the average weights of popcorns from two production lines. Experience shows that the weights of popcorns can be assumed to be normal. He would like to know whether the true mean of the weights of popcorns from two production lines are different than each other.

$$H_0: \ \mu_1 = \mu_2$$
$$H_a: \ \mu_1 \neq \mu_2$$

Find the type-II error, β, when $\mu_1 - \mu_2 = -0.3$ with $\alpha = 0.05$?



Class exercise

A grinding machine will be qualified for a particular task if it can be shown to produce less than 8% defective parts. In a random sample of 300 parts, 12 were defective. Based on this data, can the machine be qualified? To answer this question:

a) Formulate the hypothesis test.

b) Find the p-value.

c) What is your conclusion for α=1%? How about α=5%?

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